

# Indian Institute of Technology (Indian School of Mines) Dhanbad

## Winter Mid Semester Question Paper 2024-25

Examination: B.Tech (1<sup>st</sup> Year Common)

Time: 2 Hours

Discipline: Mechanical Engineering

Total Marks:30

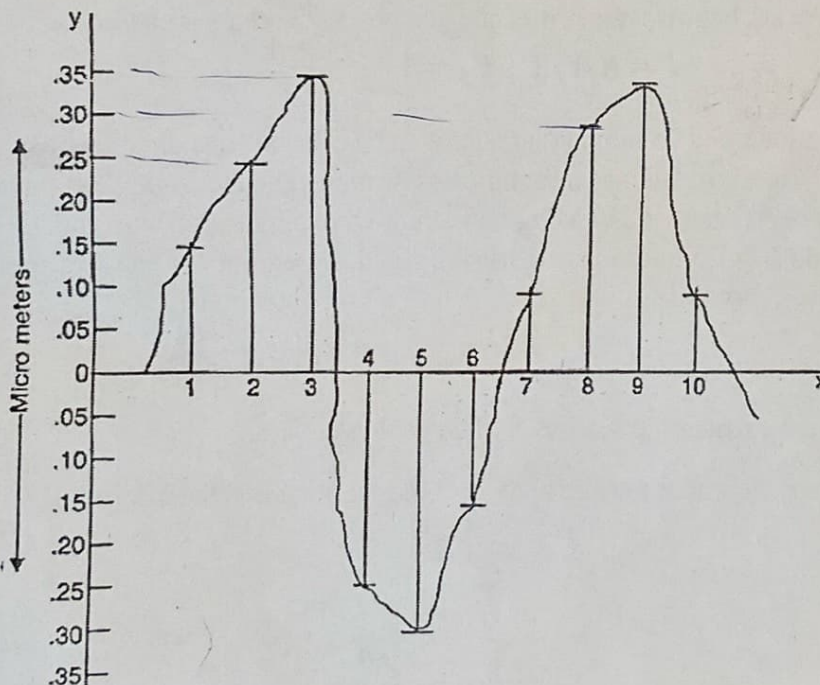
Subject: Manufacturing Process (NMES 101)

Session: 2024 - 25

(Answer all the questions. Provide the answer to each part in one continuous space without breaks)

### Part – I (Metrology – 10 marks)

1. Define the following terms: (a) Calibration, (b) Specification, (c) Inspection, (d) Accuracy, (e) Precision  
[0.4 × 5 = 2 marks]
2. Write names of following standards (a) IS 1 (b) IS 2347 (c) IS 14543 (d) IS 366 (e) ISO 9001:2015  
[0.4 × 5 = 2 marks]
3. A manufacturer produces cylindrical shafts with a nominal diameter of 50 mm. A quality control engineer measures the diameter of five randomly selected shafts using a micrometer, and the recorded measurements are 49.85, 49.90, 49.87, 49.92, 49.89 (mm). Calculate the accuracy and precision of the measurements. Comment on whether the manufacturing process is more accurate, precise or neither based on the calculated values.  
[0.75 + 0.75 + 0.5 = 2 marks]
4. Holes of  $75 \pm 0.05$  mm diameter are to be checked by the help of GO and NO-GO Plug gage. Design the GO and NO-GO Plug gages, sketch it and show its GO-size and NO-GO size dimensions. Assume normal wear allowance and gage maker's tolerance.  
[0.75 + 0.75 + 0.5 = 2 marks]
5. Calculate the  $R_a$  and the  $rms$  values for the surface roughness shown in the figure.  
[1 + 1 = 2 marks]





**Part – II (Casting– 10 marks)**

6. A cylindrical steel rod with smooth surface is to be manufactured by casting and machining. The final desired product dimension should be 140 mm in length and 46 mm in diameter. The wooden pattern has 150 mm length and 50 mm diameter. During solidification, the cast metal experiences uniform contraction of 6% in all the directions. Single stage machining is carried out to smoothen the surfaces of the cast product. Considering both machining and shrinkage allowances, calculate the thickness of material that should be removed from all directions during machining. [3 marks]
7. A cubic riser needs to be designed for a sand mould. The part to be cast is a 150 mm × 150 mm × 25 mm plate. Find the dimensions of the riser so that its total solidification time is 30% longer than the casting. [3 marks]
8. A spherical shaped casting solidifies in 20 minutes. Determine the solidification time of a spherical shaped cast of same material which is 8 times heavier. [3 marks]
9. Why misrun defect occurs in sand casting? [1 mark]

**Part – III (Forming – 10 marks)**

10. A 300-mm-wide strip 25 mm thick is fed through a rolling mill with two powered rolls each of radius 250 mm. The work thickness is to be reduced to 22 mm in one pass at a roll speed of 50 rev/min. The work material has a flow curve defined by  $K = 415$  MPa and  $n = 0.15$ , and the coefficient of friction between the rolls and the work is assumed to be 0.12. Determine if the friction is sufficient to permit the rolling operation to be accomplished. If so, calculate the roll force, torque, and horsepower. (Take 1 H.P.=745.7 W) [2 marks]
11. A cylindrical work-piece is subjected to a cold upset forging operation. The starting piece is 75 mm in height and 50 mm in diameter. It is reduced in the operation to a height of 36 mm. The work material has a flow curve defined by  $K=415$  MPa and  $n=0.17$ . Assume a coefficient of friction of 0.1. Determine the force as the process begins, at intermediate heights of 62 mm, 49 mm, and at the final height of 36 mm.  
$$F = K_f Y_f A ; K_f = 1 + \frac{0.4 \mu D}{h}$$
 [2 marks]
12. A billet 75 mm long and 25 mm in diameter is to be extruded in a direct extrusion operation with extrusion ratio  $r_x = 4.0$ . The extrudate has a round cross section. The die angle (half-angle) =  $90^\circ$ . The work metal has a strength coefficient = 350 MPa, and strain hardening exponent  $n = 0.18$ . Use the Johnson formula with  $a = 0.8$  and  $b = 1.5$  to estimate extrusion strain. Determine the pressure applied to the end of the billet as the ram moves forward.  
$$\epsilon_x = a + b \ln r_x ; p = \bar{Y}_f \left( \epsilon_x + \frac{2L}{D_0} \right)$$
 [2 marks]
13. Classify different extrusion processes with suitable sketches. [2 marks]
14. Write down the differences between hot and cold working in tabular form. [2 marks]

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